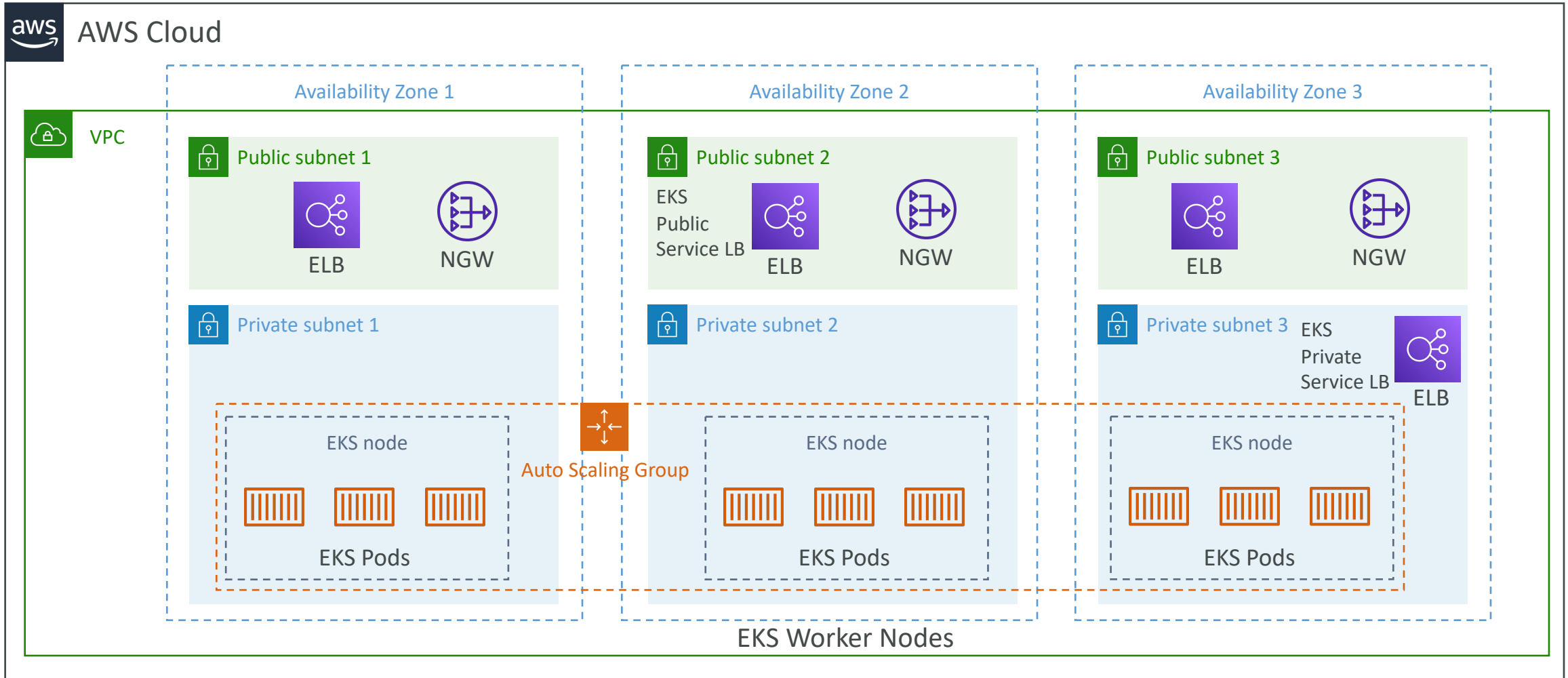




Amazon EKS Overview

- Amazon EKS = Amazon Elastic **Kubernetes** Service
- It is a way to launch **managed Kubernetes clusters on AWS**
- Kubernetes is an **open-source system** for automatic deployment, scaling and management of containerized (usually Docker) application
- It's an alternative to ECS, similar goal but different API
- EKS supports **EC2** if you want to deploy worker nodes or **Fargate** to deploy serverless containers
- **Use case:** if your company is already using Kubernetes on-premises or in another cloud, and wants to migrate to AWS using Kubernetes
- **Kubernetes is cloud-agnostic** (can be used in any cloud – Azure, GCP...)
- For multiple regions, deploy one EKS cluster per region
- Collect logs and metrics using **CloudWatch Container Insights**

Amazon EKS - Diagram



Amazon EKS – Node Types

- **Managed Node Groups**
 - Creates and manages Nodes (EC2 instances) for you
 - Nodes are part of an ASG managed by EKS
 - Supports On-Demand or Spot Instances
- **Self-Managed Nodes**
 - Nodes created by you and registered to the EKS cluster and managed by an ASG
 - You can use prebuilt AMI - Amazon EKS Optimized AMI
 - Supports On-Demand or Spot Instances
- **AWS Fargate**
 - No maintenance required; no nodes managed

Amazon EKS – Data Volumes

- Need to specify **StorageClass** manifest on your EKS cluster
- Leverages a **Container Storage Interface (CSI)** compliant driver
- Support for...
- Amazon EBS
- Amazon EFS (works with Fargate)
- Amazon FSx for Lustre
- Amazon FSx for NetApp ONTAP

